

CHRISCO DIGITAL ACADEMY

AI & TECHNOLOGY

The Practical Handbook Series

# SWE & LLM Mastery

*Build real, AI-powered software with modern engineering.*

You will learn software engineering foundations, how large language models work, prompt and API skills, and how to design, build, test and deploy an AI-powered application of your own.

8 CHAPTERS - SELF-STUDY GUIDE - PRACTICAL EXERCISES

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# SWE & LLM Mastery

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## A note before you begin

We are living through a rare shift: large language models (LLMs) have made it possible to build genuinely intelligent software faster than ever. Combine solid software engineering with the ability to harness LLMs, and you become the kind of builder companies are desperate to hire - or the founder who ships products alone.

This book assumes you can already code a little (or are learning). It focuses on how to think like an engineer and how to build with AI, not on one specific language. The principles here will outlast any single tool or model.

The goal is a working AI-powered app you build yourself. Nothing proves skill - to employers, clients, or yourself - like a real thing that works. Build as you read.

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## HOW TO USE THIS BOOK

Read one chapter at a time. Don't just read - do the 'Try This' exercise in each chapter before you move on, and keep your answers in a notebook. The 'In Practice' stories show how the idea works in real life. By the last page you should have real work to show, not just notes.

# Software engineering basics

Software engineering is more than writing code - it's building software that is reliable, maintainable and solves real problems. That means breaking problems into clear parts, writing clean and understandable code, using version control (like Git) to track changes, and thinking about how pieces fit together. These habits separate a hobbyist from a professional engineer.

Key foundations include understanding data structures and logic, reading and reusing others' code, debugging systematically, and collaborating using tools like Git and GitHub. You don't need to know everything - you need strong fundamentals and the ability to learn continuously, because technology changes fast. Good engineering thinking is the durable skill; specific tools come and go.

## IN PRACTICE

Two coders build the same feature. One writes messy code that only they understand; the other writes clean, organised code with clear names and version control. Six months later, only the second can easily fix and extend it - that's engineering, not just coding.

## TRY THIS

Set up Git and a GitHub account. Put a small piece of code (even a simple script) into a repository, making at least two commits.

## Key takeaways

- Engineering = reliable, maintainable, problem-solving code.
- Use version control (Git) from the start.
- Strong fundamentals plus continuous learning win.

## Understanding LLMs

Large language models are AI systems trained on vast amounts of text to predict and generate human-like language. They can write, summarise, translate, answer questions, and even help write code. Crucially, understand what they are: powerful pattern-predictors, not databases of facts - which is why they can be fluent yet occasionally confidently wrong ('hallucinate').

As a builder, you don't need to train these models yourself - you use existing ones (like those from OpenAI, Anthropic, Google and open-source options) via APIs. Knowing their strengths (language tasks, reasoning help, code assistance) and limits (accuracy, up-to-date facts, cost per use) lets you design smart applications that use LLMs where they shine and guard against where they fail.

### IN PRACTICE

A developer building a customer-support tool used an LLM to draft helpful replies but added a step to pull real, verified answers from the company's own documents - combining the model's fluency with reliable facts to avoid confident wrong answers.

### TRY THIS

Use an LLM (via a chat interface) for three different tasks: writing, summarising, and helping with code. Note where it excelled and where it erred.

### Key takeaways

- LLMs predict language - fluent but not always factual.
- You use existing models via APIs, not train them.
- Design around their strengths and guard their limits.

## Prompt engineering

Prompt engineering is the skill of instructing an LLM to get reliable, high-quality output - essential when building AI apps. A strong prompt provides clear context, a specific task, the desired format, and often examples. In applications, prompts are part of your code: well-crafted, tested prompts are the difference between an AI feature that delights users and one that frustrates them.

Techniques include giving the model a role, breaking complex tasks into steps, providing examples of the desired output (few-shot prompting), and instructing it to reason step by step for harder problems. You also learn to handle and validate the model's responses in code. Treat prompts as you would any code: iterate, test with many inputs, and refine until they're robust.

### IN PRACTICE

In an app that categorises customer messages, the developer's first prompt gave inconsistent results. By adding a clear role, three examples, and a strict output format, the categorisation became reliable enough to ship - the prompt itself was the fix.

### TRY THIS

Write a prompt that reliably makes an LLM output a specific format (e.g. always return JSON with certain fields). Test it on several different inputs.

### Key takeaways

- Prompts are part of your app's code.
- Use role, context, examples and clear format.
- Iterate and test prompts against many inputs.

## Building with APIs

APIs let your software talk to LLMs and other services programmatically. Instead of a person typing into a chat, your code sends a request (with your prompt and settings) to the model's API and receives a response to use in your app. Learning to authenticate, send requests, handle responses, and manage errors and rate limits is the core skill for building AI-powered software.

You'll also work with other APIs - databases, payment, external data - since real apps combine many services. Key professional concerns include keeping API keys secret (never in public code), handling failures gracefully, and being mindful of cost (LLM APIs charge per use). Mastering API integration is what lets you assemble powerful applications from existing building blocks.

### IN PRACTICE

A developer built a study-helper app where the user's question is sent via API to an LLM, and the answer is displayed in the app. The whole 'intelligence' came from an API call - he focused on the experience around it, not building a model.

### TRY THIS

Write code that sends a prompt to an LLM API and prints the response. Add basic error handling for when the request fails. Keep your API key secret.

### Key takeaways

- Your code calls LLMs and services via APIs.
- Handle auth, errors, rate limits and cost.
- Never expose API keys in public code.

## AI-powered app development

Now you combine engineering and LLMs into a real application. This means designing the app: what problem it solves, the user experience, the flow of data, and where the LLM fits. Good AI apps use the model as one component within solid engineering - with a clear interface, reliable data handling, and thoughtful prompts - rather than just a thin wrapper around a chatbot.

Think about architecture: a front end users interact with, back-end logic, data storage, and the LLM integration, all working together. Consider the user's journey and edge cases (what if the AI returns something wrong or nothing?). The best AI products feel seamless because good engineering surrounds and supports the AI, handling the messy reality the model alone can't.

### IN PRACTICE

A team built an app that turns a user's rough notes into organised meeting minutes: the front end captures notes, the back end sends a carefully-prompted request to an LLM, validates the result, and saves it. Solid engineering around the AI made it genuinely useful.

### TRY THIS

Design (on paper) an AI-powered app: the problem it solves, the user flow, where the LLM fits, and how you'll handle wrong or empty AI responses.

### Key takeaways

- Use the LLM as one component of solid engineering.
- Design the full app: front end, logic, data, AI.
- Handle edge cases and wrong AI outputs.

## Testing and deployment

Software isn't done when it works on your machine - it must work reliably for users. Testing means checking that your app behaves correctly, including with unexpected inputs and when the AI misbehaves. Write tests for your logic, and manually test the whole user experience. With AI features, also test that your prompts and validation handle a wide range of inputs gracefully.

Deployment means putting your app where users can access it - on a hosting platform or cloud service. You'll learn to deploy your application, manage environment settings (like API keys securely), and monitor it once live. Shipping a real, deployed app - not just code on your laptop - is a major milestone and exactly what proves professional capability.

### IN PRACTICE

A developer's app worked perfectly in testing until a user entered something unexpected and the AI returned nonsense. Because he'd added validation and tested edge cases before deploying, the app handled it gracefully instead of crashing in front of real users.

### TRY THIS

Test one part of your app with unusual and empty inputs and fix what breaks. Then deploy a simple app to a free hosting platform so it's live online.

### Key takeaways

- Test logic, user experience, and AI edge cases.
- Deploy so real users can access it.
- Keep secrets safe and monitor the live app.

## Real-world projects

Skill becomes credibility through real projects. Build things that solve genuine problems - for yourself, a community, or a small business - and put them on GitHub as your portfolio. Employers and clients care far more about what you've actually built than about certificates. A few solid, well-documented projects demonstrating engineering plus AI integration speak louder than any CV line.

Work on projects end-to-end: from idea and design through building, testing, deploying, and writing a clear README explaining what it does and how. Contributing to open-source or collaborating with others also builds real-world experience and network. Each project teaches new problems and grows your ability - and your body of work becomes your strongest career asset.

### IN PRACTICE

A self-taught developer landed a job not through a degree but through three deployed projects on GitHub - including an AI-powered tool - each with a clear README. The interviewer said the working, documented projects made the decision easy.

### TRY THIS

Plan a portfolio project that solves a real problem using engineering + an LLM. Build it end-to-end and write a clear README for it on GitHub.

### Key takeaways

- Real, deployed projects prove skill best.
- Document projects clearly on GitHub.
- Your body of work is your strongest asset.

## Build your AI app

Your capstone is a complete, working AI-powered application that you design, build, test and deploy yourself. Choose a real problem an LLM can genuinely help solve, keep the scope achievable, and take it all the way to a live, usable product. This ties together every skill in the book - engineering fundamentals, LLM understanding, prompting, APIs, app design, testing and deployment.

Expect challenges - integrating everything and handling real-world messiness is hard, and that struggle is exactly where mastery forms. Break it into steps, use documentation and AI assistants when stuck, and keep going until it runs for real users. When it's live, you'll have crossed from 'learning' to 'builder of AI software' - with proof in hand and confidence earned.

### IN PRACTICE

A learner built and deployed a small app that summarises long documents into key points using an LLM API, with a clean interface and error handling. It was modest but real, live, and useful - and it became the project that opened doors for her.

### TRY THIS

Design, build, test and deploy one complete AI-powered app this course. Take it all the way to live, then share the link and code as proof.

### Key takeaways

- Ship one complete, live AI-powered app.
- The struggle of integration builds mastery.
- A deployed app is proof you're an AI builder.



KEEP GOING

# Skills pay for life. Now go use them.

Finishing the reading is not the goal - finishing the project is. Pick the one exercise in this book that scared you most and do it this week. Then bring your work to your CHRISCO facilitator or post it online. That single habit - shipping real work - is what separates the youth who earn from the youth who only dream.

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